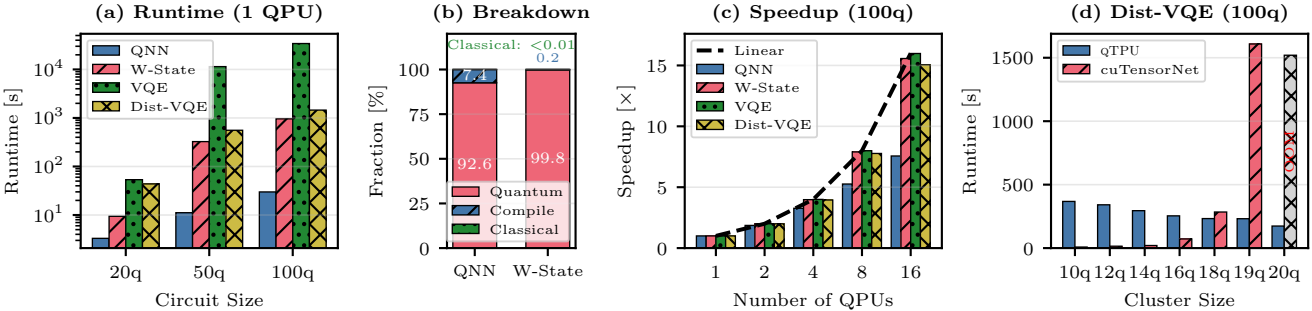
(a) Runtime (1 QPU)
QNN
W-State
VQE
Dist-VQE
Runtime [s]
$10^4$
$10^3$
$10^2$
$10^1$
20q
50q
100q
Circuit Size
(b) Breakdown
Classical: <0.01
0.2
7.4
92.6
99.8
Fraction [%]
100
75
50
25
0
Quantum
Compile
Classical
QNN
W-State
(c) Speedup (100q)
Linear
QNN
W-State
VQE
Dist-VQE
Speedup [×]
15
10
5
0
1
2
4
8
16
Number of QPUs
(d) Dist-VQE (100q)
qTPU
cuTensorNet
Runtime [s]
1500
1000
500
0
OOM
10q
12q
14q
16q
18q
19q
20q
Cluster Size